SIMPÓSIO DE EFICIÊNCIA ENERGÉTICA, EMISSÕES E COMBUSTÍVEIS

O programa VEIN: mapeamento de emissões veiculares em condições de trafego urbano real

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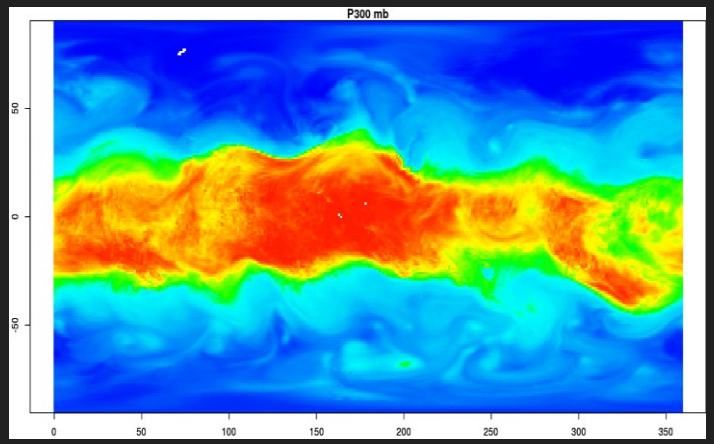
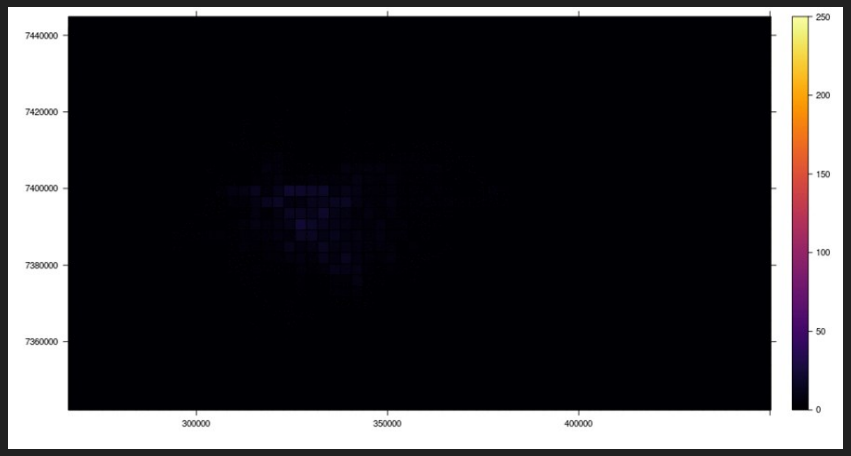
VEIN

Air quality simulations

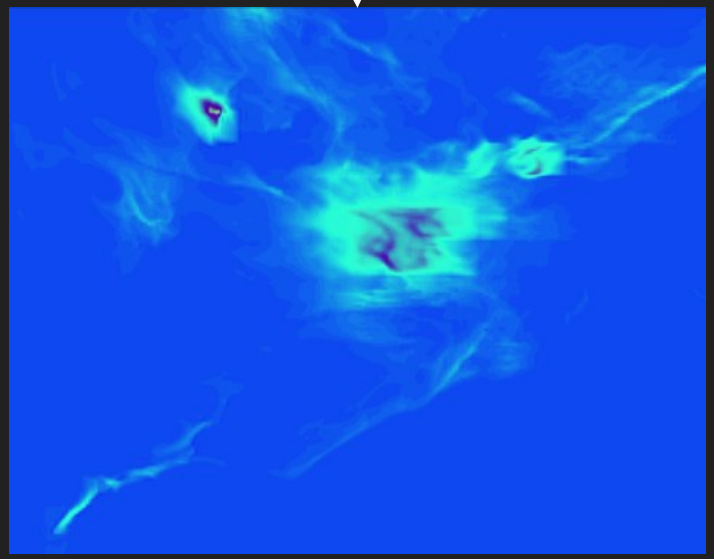
Meteorology



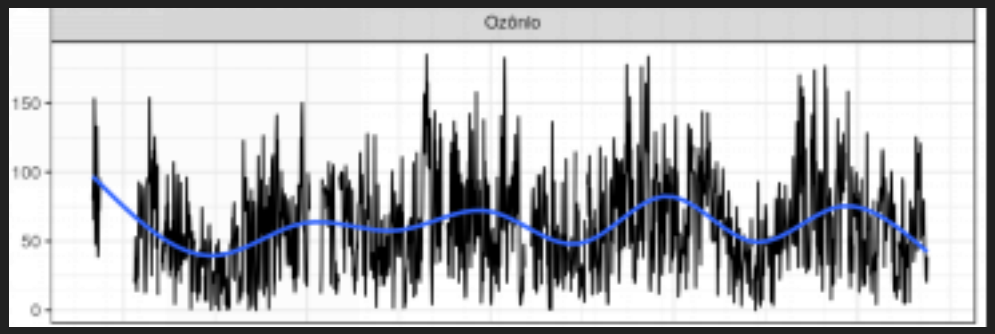


Emissions

Air quality model



Air Quality Assessment



The problem:
Air quality simulations, ‘eventually’, do match with air quality observations

The emission inventories are easily seen as the scapegoat if a mismatch is found between modelled and observed concentrations of air pollutants (Pulles and Helsinga, 2003).

https://CRAN.R-project.org/package=vein

https://www.geosci-model-dev-discuss.net/gmd-2017-193/

vein: Vehicular Emissions Inventories

Elaboration of vehicular emissions inventories, consisting in four stages, pre-processing activity data, preparing emissions factors, estimating the emissions and post-processing of emissions in maps and databases.

Version:	0.3.9
Depends:	R (≥ 2.10)
Imports:	sf, sp, data.table, graphics, stats, units, methods
Suggests:	knitr, rmarkdown, testthat, covr
Published:	2018-03-01
Author:	Sergio Ibarra-Espinosa [aut, cre]
Maintainer:	Sergio Ibarra-Espinosa <sergio.ibarra at usp.br>
BugReports:	https://github.com/atmoschem/vein/issues/
License:	MIT + file LICENSE

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VEIN v0.2.2: an R package for bottom–up vehicular emissions inventories

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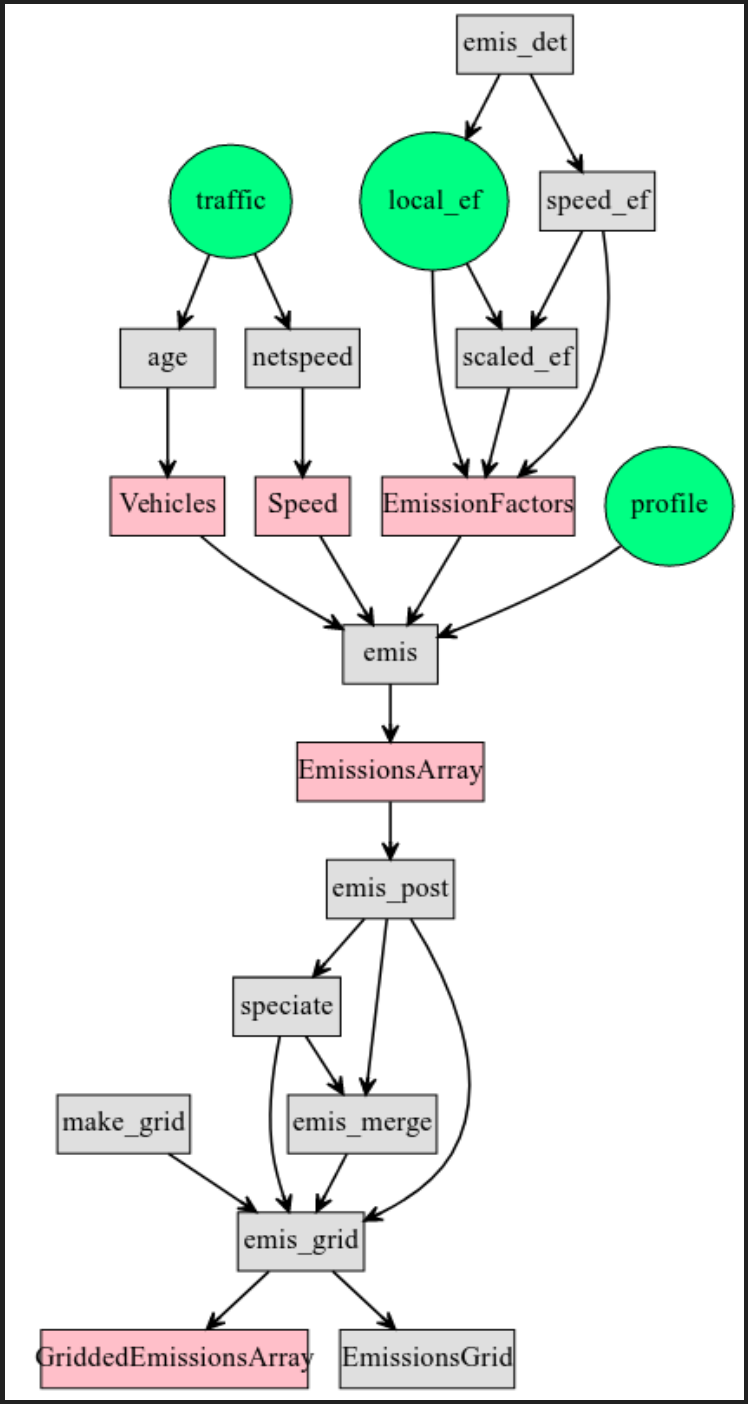
Intro

- Pollution caused 9 million premature deaths in 2015, 16% of all deaths worldwide (LANDRIGAN et al, 2017).
- Air pollution also affects ecosystemic services.
- There is consensus that climate change is due to human activity (COOK et al., 2016).
- It is crucial to understand the physical and chemical mechanism of generation of the air pollutants, the emissions.
- In urban centers, vehicles are the most important source of air pollution (MOLINA; MOLINA, 2004b).

VEIN

- Vehicular Emission INventory is an R package to estimate vehicular emissions https://CRAN.R-project.org/package=vein
- The manuscript describing the core functionalities of VEIN is currently unde review on GMDD https://www.geosci-model-dev-discuss.net/gmd-2017-193/ (v0.2.2)
- This figure shows the process update including classes (pink), functions (grey) and data (green)



Installation

System Requirements

vein (https://atmoschem.github.io/vein/) imports function of sf (https://github.com/r-spatial/sf), hence, the user must install its requisites. In Ubuntu:

```
sudo add-apt-repository ppa:ubuntugis/ubuntugis-unstable
sudo apt-get update && sudo apt-get install libudunits2-dev libgdal-dev
libgeos-dev libproj-dev libnetcdf-dev
```

Then, install the CRAN version

```
install.packages("vein")
```

or the development version on GitHub

```
devtools::install_github("atmoschem/vein")
```

New features: `inventory`

- The function inventory creates a structure of directories and scripts to help new user to run VEIN.
- The argument name it is for name of the city.
- Includes argument for creating an structure only 1 hour or several hours.
- Actually, VEIN can be used either with bottom-up or top-down approach.

```
library(vein)
inventory(name = file.path(tempdir(), "YourCity"),
          show.dir = T, show.scripts = T)

## files at /tmp/RtmpNb0Y7T/YourCity

## Directories:
##  [1] "/tmp/RtmpNb0Y7T/YourCity"
##  [2] "/tmp/RtmpNb0Y7T/YourCity/ef"
##  [3] "/tmp/RtmpNb0Y7T/YourCity/emi"
##  [4] "/tmp/RtmpNb0Y7T/YourCity/emi/BUS_01"
##  [5] "/tmp/RtmpNb0Y7T/YourCity/emi/HGV_01"
##  [6] "/tmp/RtmpNb0Y7T/YourCity/emi/LCV_01"
##  [7] "/tmp/RtmpNb0Y7T/YourCity/emi/MC_01"
##  [8] "/tmp/RtmpNb0Y7T/YourCity/emi/PC_01"
##  [9] "/tmp/RtmpNb0Y7T/YourCity/est"
## [10] "/tmp/RtmpNb0Y7T/YourCity/images"
## [11] "/tmp/RtmpNb0Y7T/YourCity/network"
## [12] "/tmp/RtmpNb0Y7T/YourCity/post"
## [13] "/tmp/RtmpNb0Y7T/YourCity/post/df"
## [14] "/tmp/RtmpNb0Y7T/YourCity/post/grids"
## [15] "/tmp/RtmpNb0Y7T/YourCity/post/streets"
## [16] "/tmp/RtmpNb0Y7T/YourCity/profiles"
## [17] "/tmp/RtmpNb0Y7T/YourCity/veh"
## Scripts:
## [1] "est/BUS_01_input.R" "est/HGV_01_input.R" "est/LCV_01_input.R"
## [4] "est/MC_01_input.R"  "est/PC_01_input.R"  "main.R"
## [7] "post.R"             "traffic.R"
```

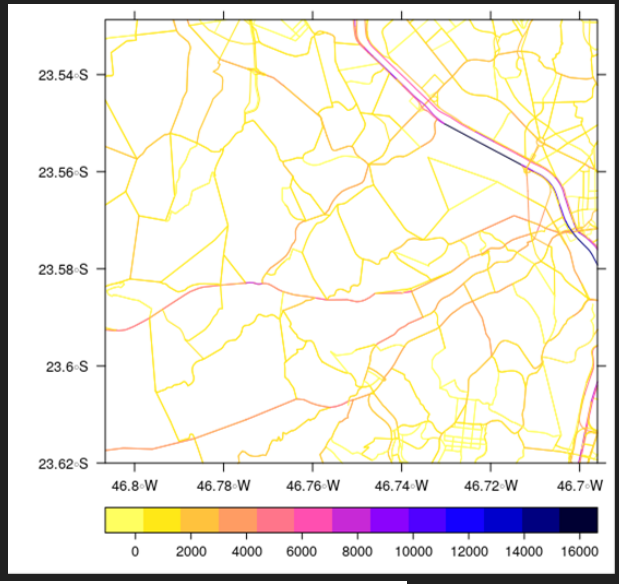
Classes

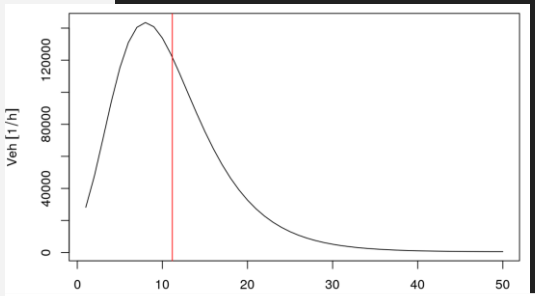
1. Vehicles (1/h).
2. Speed (km/h).
3. EmissionFactorsList (g/km, list of speed functions).
4. EmissionFactors (g/km).
5. Emissions (g/h).
6. EmissionsArray (g/h, units not explicit).
7. EmissionsList (g/h).
8. Evaporative (1/d)
9. GriddedEmissionsArray (g/h)

using vein: 1) Traffic, network and speed

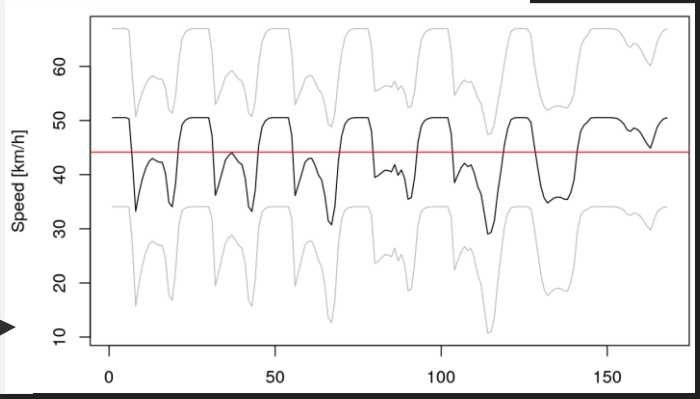
Get traffic data or use example

```
library(vein)              # vein
data("net")                # Data
net$ldv <- as.numeric(net$ldv) # Convert to numeric
sp::spplot(net, "ldv", scales = list(draw = T))
PC_G <- age_ldv(           # Function
x = net$ldv,               # Vehicles/street
name = "PC_G",             # Name
a = 1.6, b = -0.2,         # Scrapping param
k = 1)                     # % Vehicular composition
plot(PC_G, xlab = "Age of use")
```

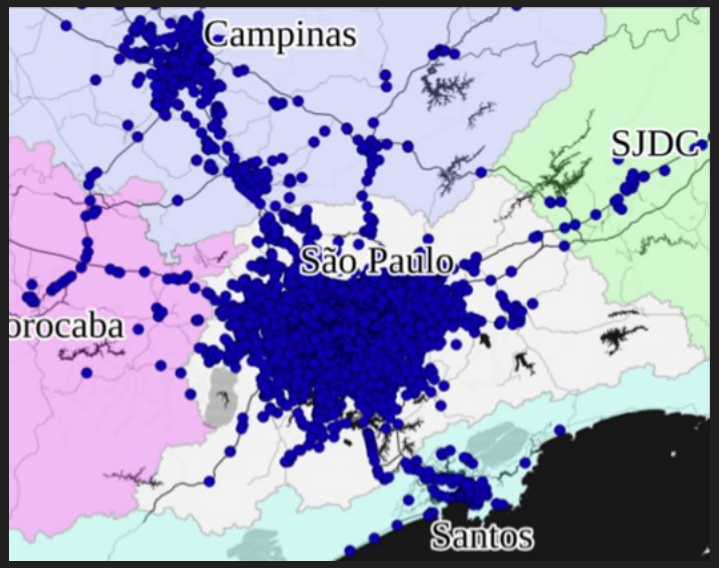




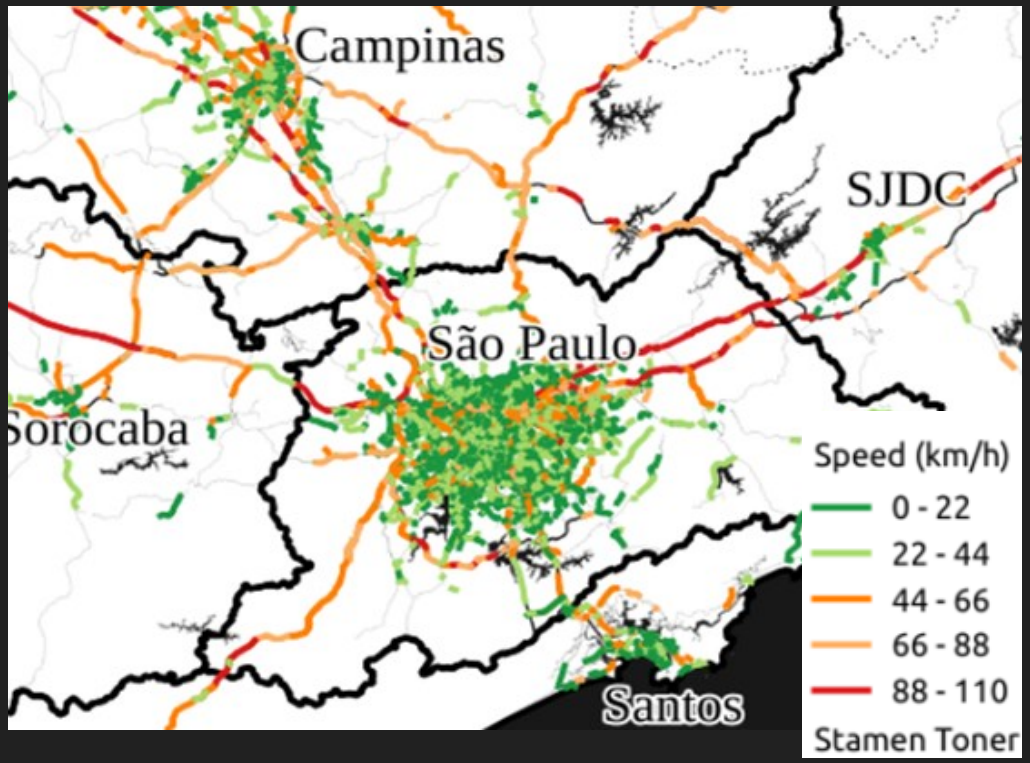

```
data("pc_profile")    # Data
pc_week <- temp_fact(# Function
net$ldv+net$hdv,      # veh/street
pc_profile)           # Temporal factor
dfspeed <- netspeed( # Function
pc_week,              # Hourly veh/street
net$ps,               #Peak speed
net$ffs,              # Free speed
net$capacity,         # Capacity
net$lkm,              # lkm
alpha = 1.5)          #BPR param
plot(dfspeed, ylab = "Speed [km/h]")
```

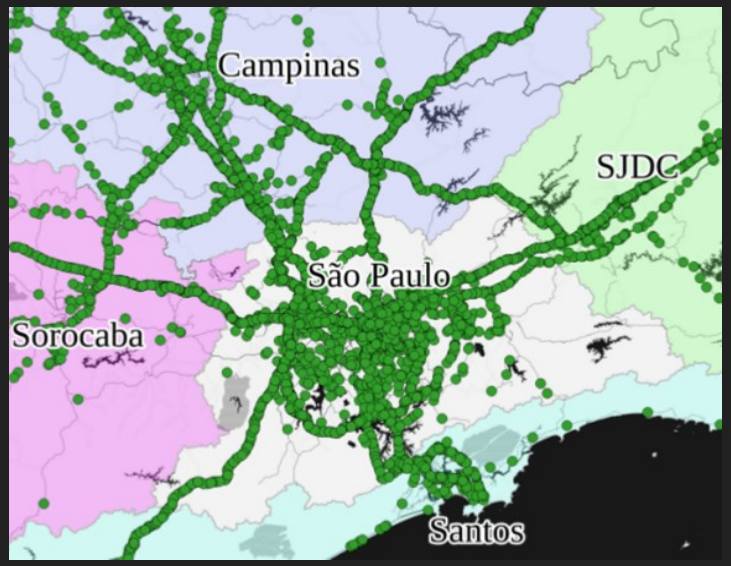


Real Traffic: GPS recordings (data from MapLink)



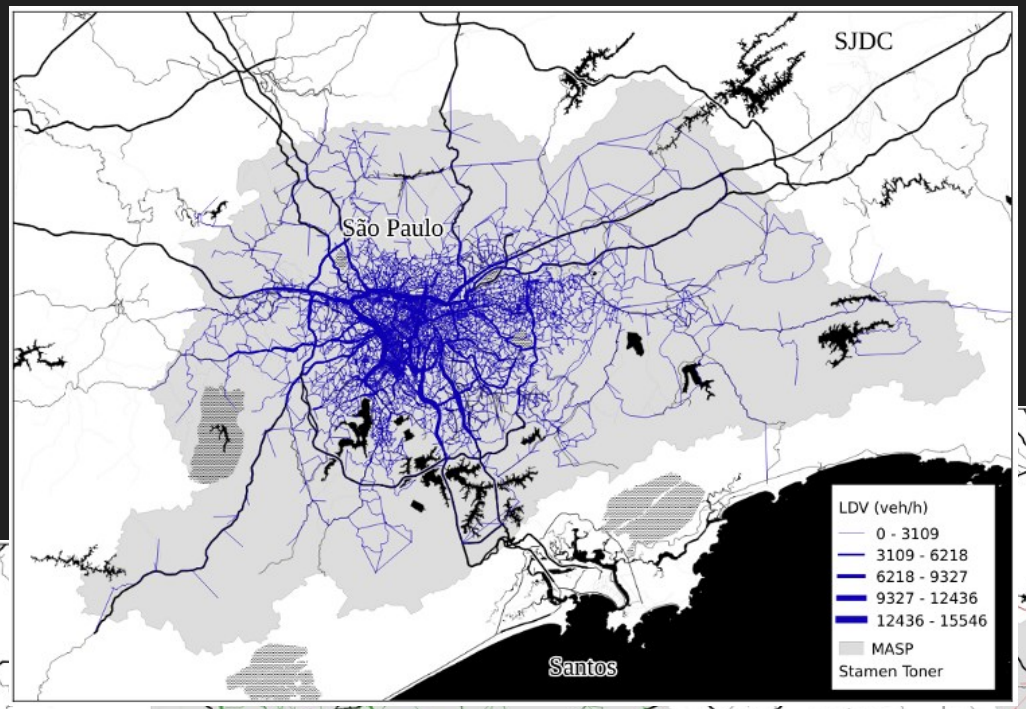
LDV





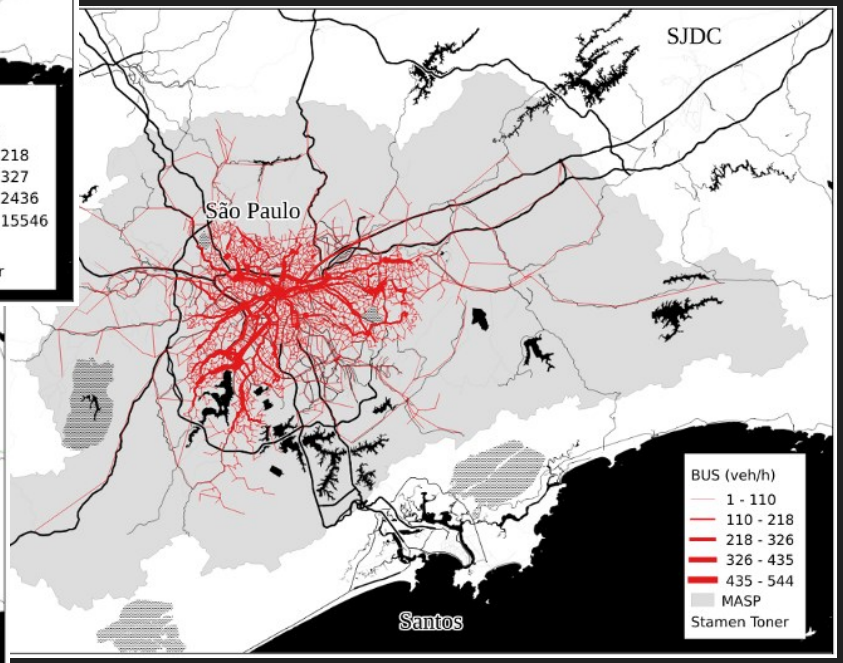
HGV

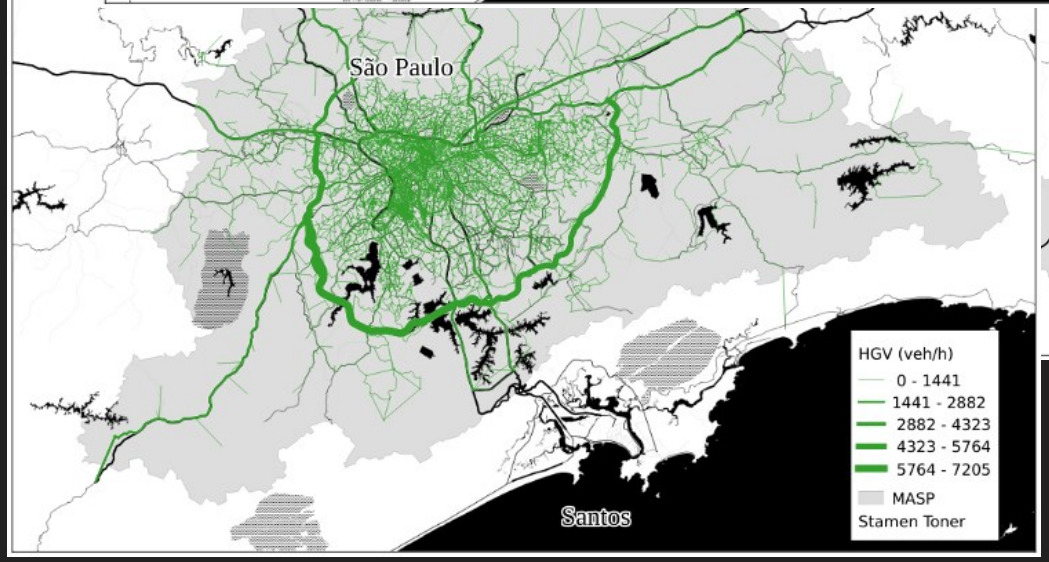
Simulated from CET and SPTRANS



LDV

BUS



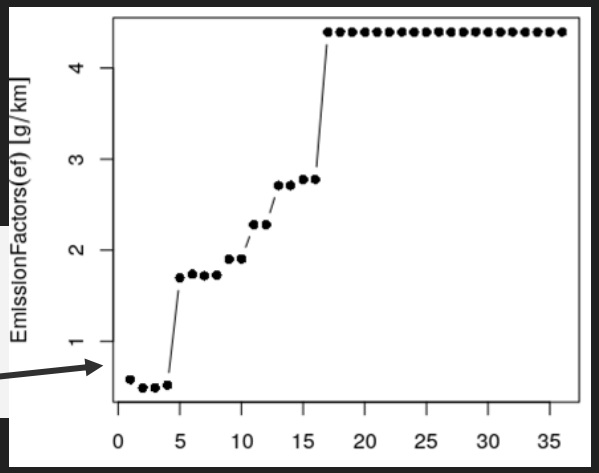


HGV

2) Select emission factors

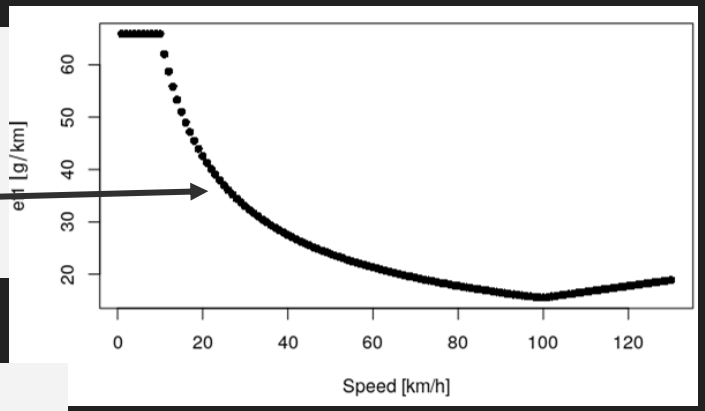
By age of use (or your local EF)

```
data("fe2015")
ef <- fe2015[fe2015$Pollutant == "NOx", # Select rows
             ]$LT                       # Select columns
plot(EmissionFactors(ef), type = "b", pch = 16)
```



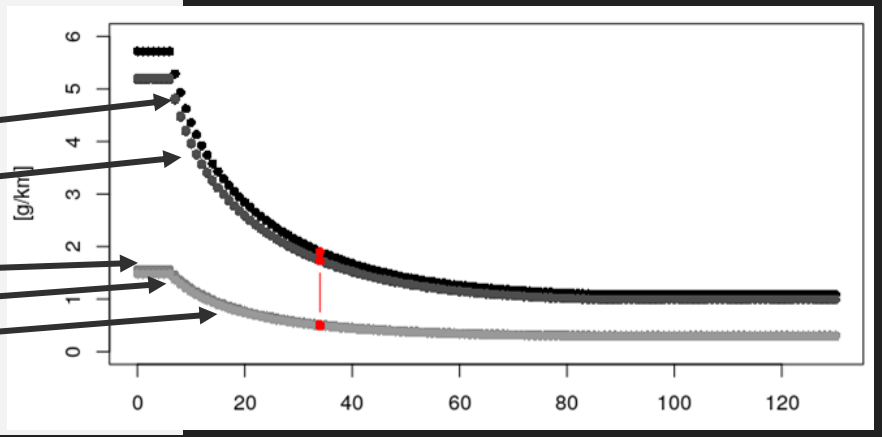
Copert speed functions

```
efp <- ef_ldv_speed(v = "PC", cc = "<=1400", f = "G",
                    eu = "PRE", p = "CO")
ef1 <- EmissionFactors(efp(1:130))
plot(Speed(1:130), ef1, xlab = "Speed [km/h]",
     type = "b", pch = 16)
```



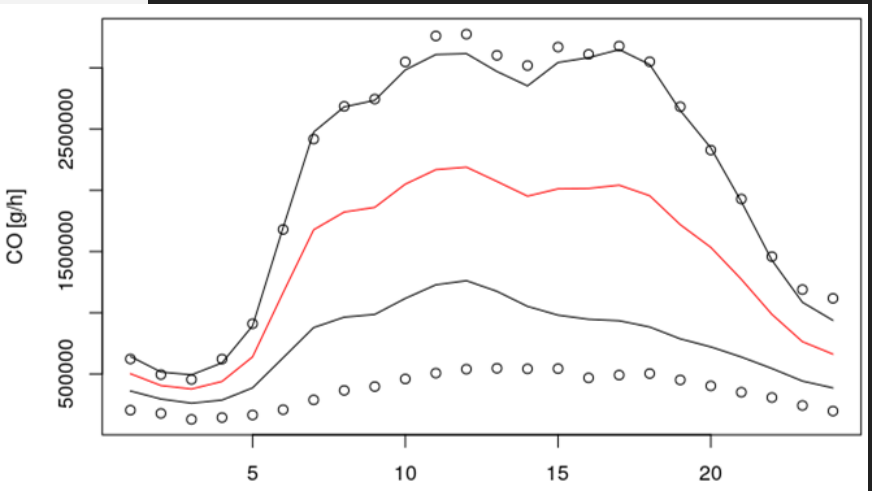
Scaled speed functions

```
eu <- fe2015[fe2015$Pollutant == "NOx", ]$Euro_HDV
lef <- ef_hdv_scaled(dfcol = ef, v = "Trucks",
                     t = "RT", g = "<=7.5", eu = eu,
                     gr = 0, l = 0.5, p = "CO")
sapply(lef, class)
 [1] "function"
 [2] "function"
 [3] "function"
 [4] "function"
 [5] "function"
```



3) Emission estimation: `emis` functions

```
data(profiles) # NEW
# just repeating peak speed. Better use netspeed!
speed <- do.call("cbind", lapply(1:168, function(i) net$ps))
E_CO <- emis(veh = PC_G,
             lkm = net$lkm,
             ef = lef,
             speed = speed,
             profile = profiles$HGV_JUNE_2014)
plot(E_CO, ylab = "CO [g/h]")
```



Quick estimation for 1 hour

```
pro1h <- matrix(1)
E_CO <- emis(veh = PC_G,  lkm = net$lkm,  ef = lef,
             speed = net$ps, profile = pro1h)
plot(E_CO, ylab = "CO [g/h]")
```

This allow quick scenario evaluationl

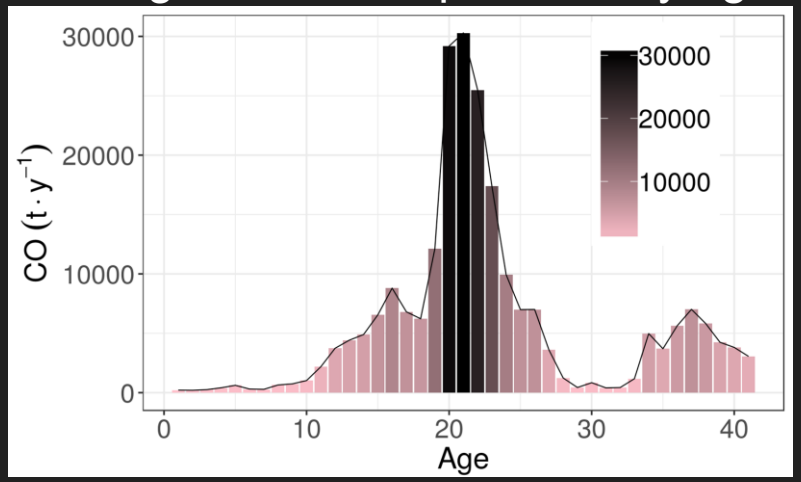
4) Post estimation: `emis_post, by = "veh"`

`emis_post` can generates two type objects, a data.frame with the sum of emissions by vehicle characteristic and the total emissions by spatial feature (see https://cran.r-project.org/web/packages/sf/vignettes/sf1.html). sp objects are converted to sf. `emis_post` can process any `EmissionsArray` with length of dimensions equal to 3 (streets, veh, hours) or 4 (streets, veh, hours, days).

Options of argument by: "veh", "streets_narrow", "streets_wide".

```
E_CO_DF <- emis_post(arra = E_CO,       # EmissionsArray
                     veh = "PC",        # Character
                     size = "<1400",    # Character
                     fuel = "G",        # Character
                     pollutant = "CO",  # Character
                     by = "veh")        # by veh!
head(E_CO_DF)
  E_CO             g veh  size fuel pollutant age hour
1 E_CO_1  2386.091 g/h  PC <1400    G        CO   1    1
2 E_CO_2  3157.963 g/h  PC <1400    G        CO   2    1
3 E_CO_3  4427.347 g/h  PC <1400    G        CO   3    1
4 E_CO_4  5981.102 g/h  PC <1400    G        CO   4    1
5 E_CO_5 22788.893 g/h  PC <1400    G        CO   5    1
6 E_CO_6 25715.426 g/h  PC <1400    G        CO   6    1
```

`by = "veh"` allow to answer an interesting question… What are the vehicles that are emitting most of the pollution by age of use?

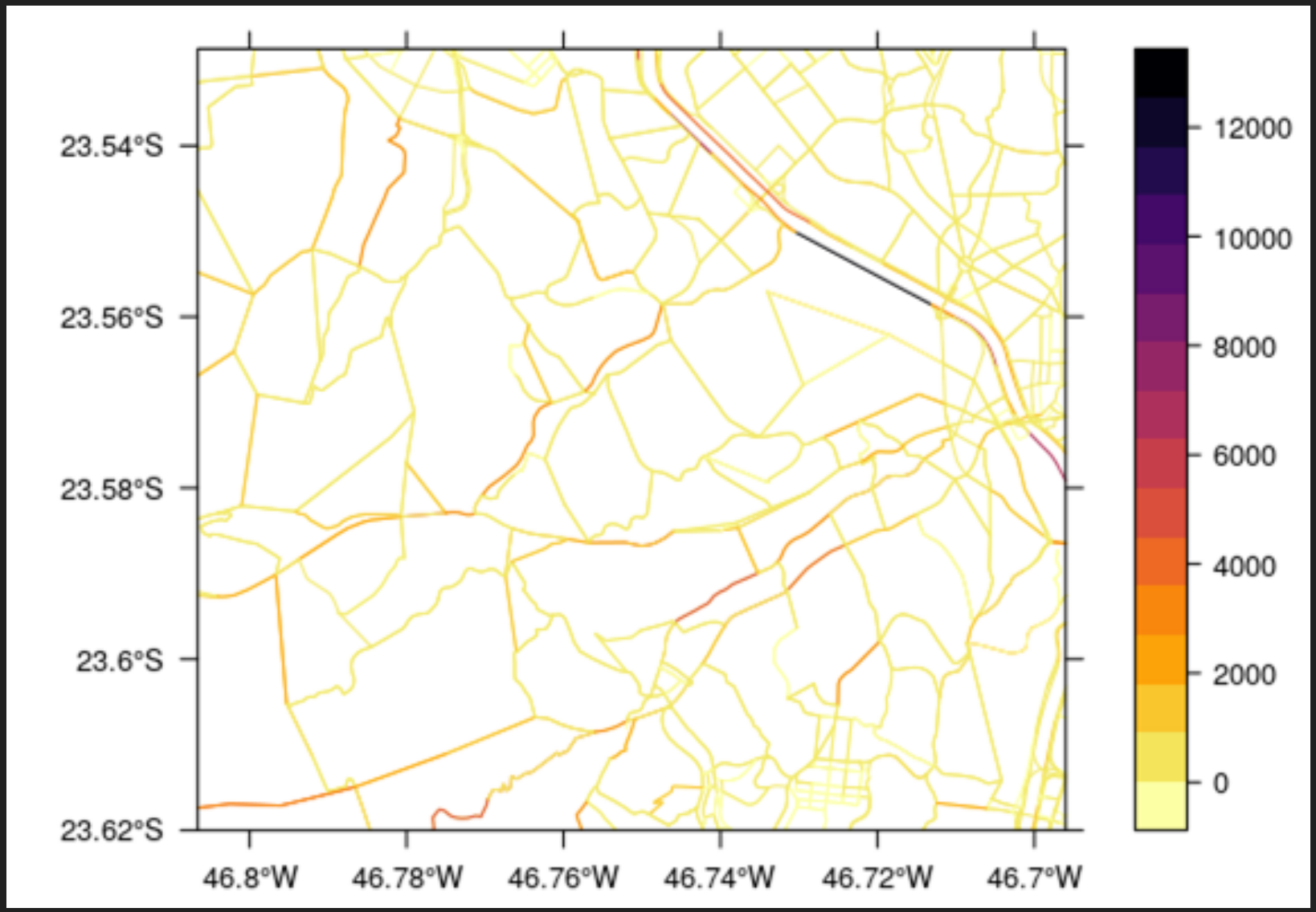


4) Post estimation: `emis_post, by = "streets_wide"`

`emis_post` with argument `by = "streets_wide"` allows to know the total amount of emissions by spatial feature, usually "LINESTRING" or street.

```
E_CO_STREETS <- emis_post(arra = E_CO,          # EmissionsArray
                          pollutant = "CO",     # Character
                          by = "streets_wide")  # Character
                                                                                >
head(E_CO_STREETS[, 1:3])                       # only first 3 hours
Result for Emissions
      V1               V2               V3
1 11709.067785 g/h 1330.353901 g/h 1640.210715 g/h
2   295.625429 g/h  230.117522 g/h  283.714901 g/h
3    48.401176 g/h   37.675916 g/h   46.451129 g/h
4    66.442525 g/h   51.719465 g/h   63.765605 g/h
5     3.131599 g/h    2.437665 g/h    3.005429 g/h
6   222.671143 g/h  173.329242 g/h  213.699888 g/h
E_CO_STREETS <- st_sf(E_CO_STREETS, geometry = net$geometry)
E_CO_STREETS <- st_sf(E_CO_STREETS,
                       geometry = st_as_sf(net)$geometry)
net@data <- cbind(E_CO_STREETS)                 # S4 for spplot only
net$V1 <- as.numeric(net$V1 )
sp::spplot(net, "V1", scales = list(draw = T),
           col.regions = rev(cptcity::cpt()))
```

Emissions test case (CO g/h)



4) Post estimation: `emis_merge`

`emis_merge` search and merge `emis_post` files

`inventory` creates a script `main.R` and `post.R` which, when ran, stores emissions outputs in the sub-directory of emi for each member of the vehicular composition.

Result: Many files.

Hence, `emis_merge` reads and merges automatically several files.

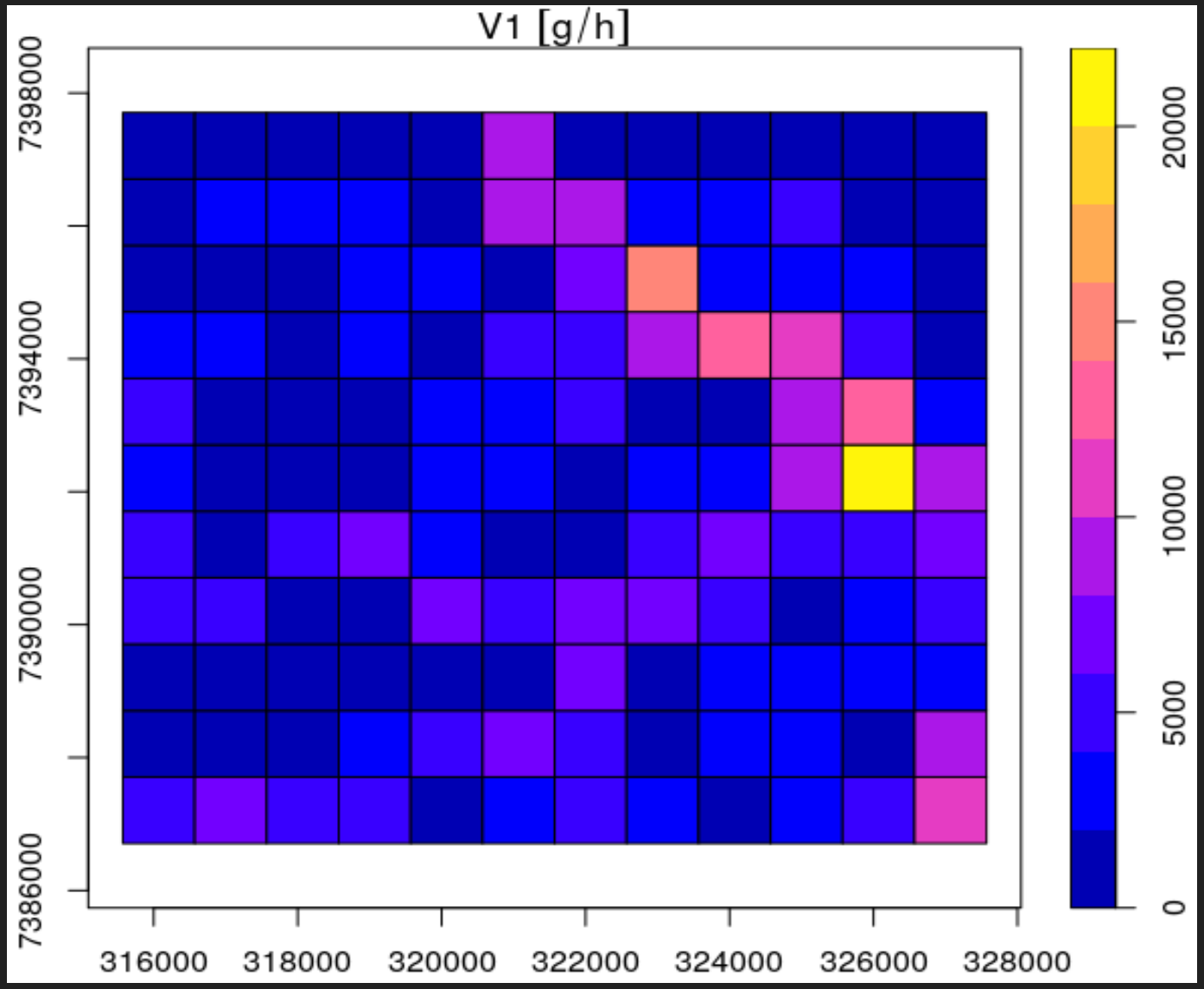
```
CO <- emis_merge('CO', net = net) # Street emissions
Reading emissions from:
[1] "CO_BUS_01_STREETS.rds" "CO_HGV_01_STREETS.rds" "CO_LCV_01_STREETS.rds"
[4] "CO_MC_01_STREETS.rds" "CO_PC_01_STREETS.rds"

dfCO <- emis_merge('CO', what = 'DF.rds', FALSE)
Reading emissions from:
[1] "CO_BUS_01_DF.rds" "CO_HGV_01_DF.rds" "CO_LCV_01_DF.rds"
[4] "CO_MC_01_DF.rds"  "CO_PC_01_DF.rds"
```

4) Post estimation: `emis_grid`

- vein imports sf functions. Older emis_grid in cities could take hours. I started to replicate `emis_grid` in QGIS taking 10 minutes. After sf integrated spatial indexes (https://www.r-spatial.org/r/2017/06/22/spatial-index.html) emis_grid took only 5.5 seconds. Therefore, migrating to sf was very significative.
- R packages `raster`, `rgdal` and `rgeos` not needed anymore.
- vein now imports data-table.
- `data.table` imported in `emis_grid`. This made the aggregation by street even faster!

```
g <- make_grid(net, 1/102.47/2, 1/102.47/2) #500m in degrees
net@data <- net@data[,- c(1:9)]
names(net)
E_CO_g <- emis_grid(spobj = net, g = g, sr= 31983)
head(E_CO_g) #class sf
gr <- GriddedEmissionsArray(E_CO_g, rows = 19, cols = 23,
                            times = 168, T)
plot(gr, col = cptcity::cpt())
```

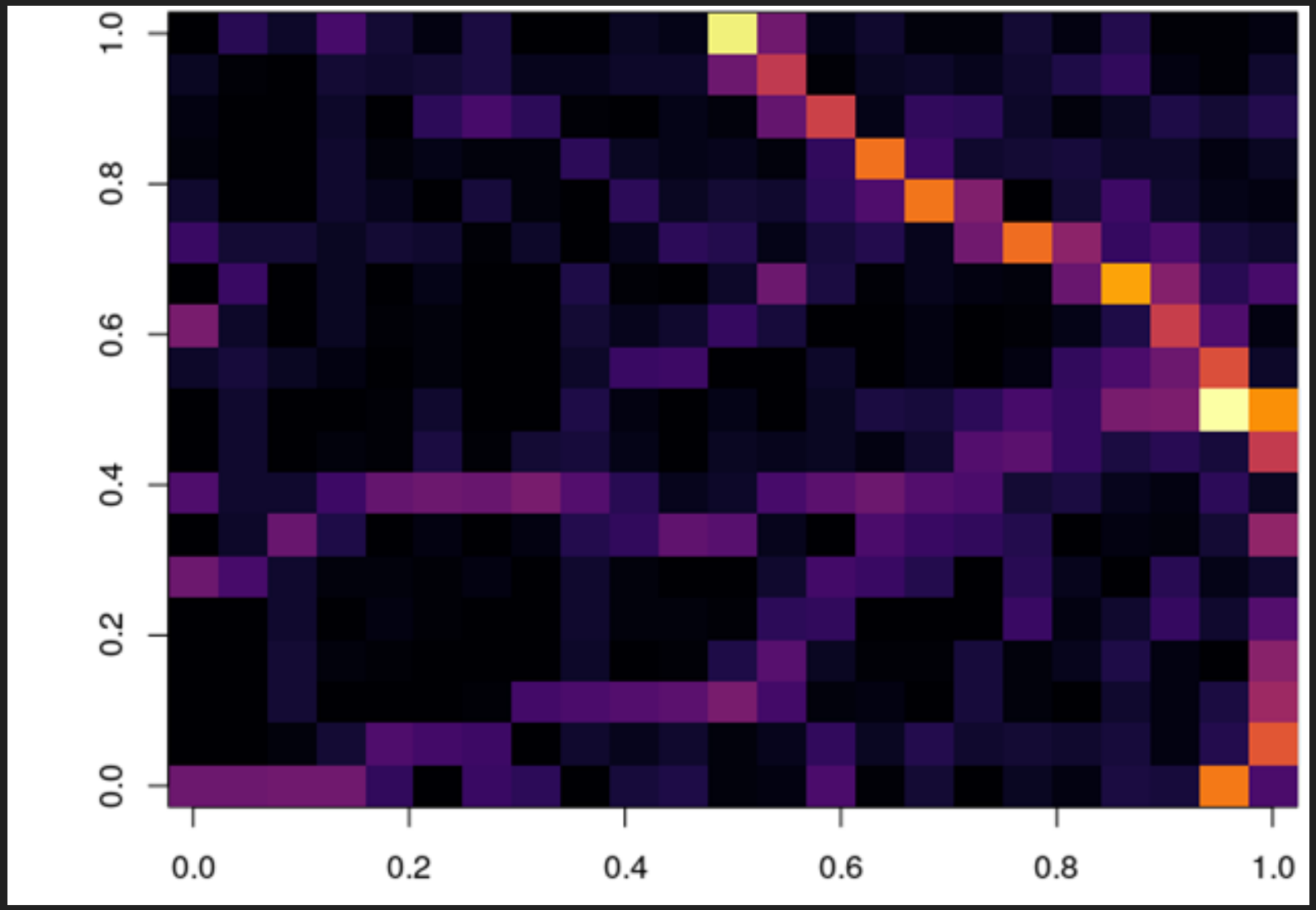
Emissions test case (CO g/h)



`GriddedEmissionsArray`.

- Add class `GriddedEmissionsArray`.
- This class read data.frames, matrix, SpatialPolygonDataFrames or sf of MULTIPOLYGON of gridded emissions to create an array of gridded data which dimensions are centroids of each cell **lat**, **lon** and **hours**.

GriddedEmissionsArray



Why `GriddedEmissionsArray`?

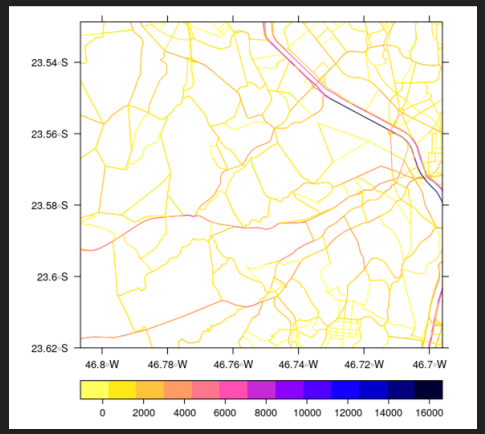
`GriddedEmissionsArray` can connect with `eixport` (Ibarra *et al*, 2018) to generates WRFchem (Grell *et al*, 2005) directly to hard drive, importing `ncdf4`, directly in R. In contrast, `emis_wrf` created an input to AAS4WRF (Vara *et al*, 2016) which also needed NCL.

The steps are:

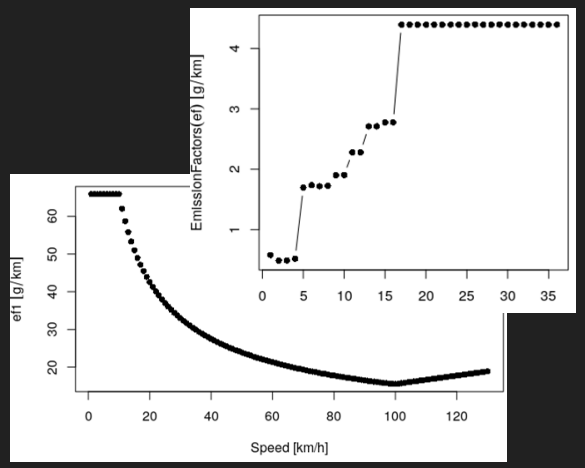
1. create a grid with `eixport::wrf_grid`,
2. create an WRFchem input with `eixport::wrf_create`
3. put the GriddedEmissionsArray in it with `eixport::wrf_put`

VEIN

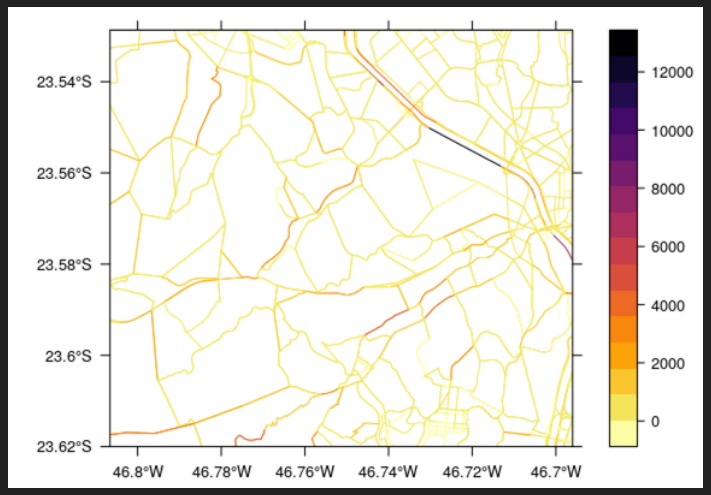
Traffic (veh/h)



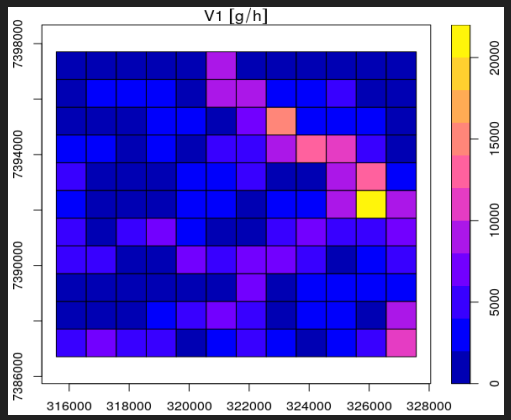
EF (g/km)



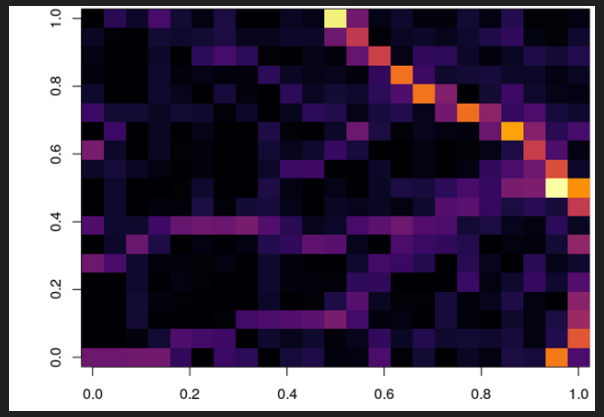
Emissions (g/h)



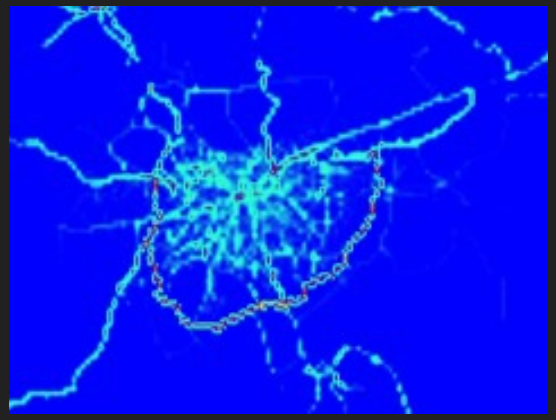
Gridded Emissions (g/h)



Array Emissions (g/h)



Emissions (mol/h)



eixport

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Thanks!

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https://atmoschem.github.io/veinreport
https://github.com/ibarraespinosa/cptcity

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